ROWNAA

Book a car, anywhere you're headed.

DEVELOPER SPECIFICATION DOCUMENT

Car Rental Marketplace Platform — Full Build Spece

1. Project Overview

Rownaa is a car rental marketplace website, structured similarly to ride-sharing platforms (e.g. Pathao, Uber) but built around scheduled rentals rather than point-to-point pickups.

The platform owner does not own any vehicles. Car owners and drivers register on the platform and list their vehicles. Customers browse and book based on pickup location, date, and service type. The platform takes a commission on each completed booking.

1.1 Business Model

- Marketplace model: car owners/drivers register and list vehicles; the platform does not own any cars
- Commission: approximately 10% per booking, taken from the total fare
- Payment split: customer pays a 20% advance online to confirm the booking; the remaining balance is paid directly to the driver in person
- Owners/drivers are verified (documents checked) before being approved to receive bookings
- Initial services offered: Airport Transfer, Corporate Transport, Car Rental with Hourly, Intercity Travel, Travel & Tour, Event & Wedding Transport.

1.2 User Roles

- Customer — books a car for a service
- Car Owner / Driver — registers a vehicle (and optionally a driver) and fulfills bookings
- Admin — verifies users, oversees bookings, tracks commission, manages disputes and platform content

2. Customer Panel — Booking Flow

The customer selects one of 6 services on the homepage, then a service-wise booking form appears.

2.1 Common Fields (present on every service)

- Name
- Mobile number
- Pickup location (address / area select — map picker preferred)
- Pickup date
- Pickup time
- Car type preference (Sedan / SUV / Microbus / Noah-Hiace / Any)
- Special note / instruction (optional text box)
- Advance payment (20%) — via bKash / Bank Transfer or SSLCommerz at confirmation

2.2 Service-Specific Fields

#	Service	Service-Specific Fields
01	Car Rental with Hourly	• Rental duration (hours / days) • Start date & time • End date & time (or "estimated") • Trip type: within city / outside city
02	Airport Transfer	• Pickup type: from Airport / to Airport • Trip type: One-way / Round trip • Drop location (if from airport) or pickup location (if to airport)
03	Intercity Travel	• From city • To city • Trip type: One-way / Round trip • Return date & time (if round trip) • Preferred pickup point (within origin city) • Drop point (within destination city)
04	Travel & Tour Booking	• Trip type: Tour (multiple destinations) • From city • To city / Destination(s) • One-way / Round trip • Trip duration (days, for tour packages) • Itinerary detail (optional — places to visit)
05	Event & Wedding Transport	• Event type (wedding / function / corporate event / other) • Event date • Venue location • Multiple cars needed? (based on guest count) • Number of cars needed (if multiple) • Duration (how many hours the car is needed)
06	Corporate Transport	• Company name • Office pickup point • Drop location • One-time / Recurring (daily / weekly / monthly) booking • If recurring: for how many days, which days • Number of passengers

2.3 Seat Matching Logic

Based on how many passengers the trip creator wants to carry, the system should automatically filter and suggest available cars matching that count.

The system should display:

- Total passengers required
- How many cars are needed
- Which car type(s) are needed
- Available seats per car
- Combined total seats across the suggested cars

Example — 10 passengers:

- Sedan — 4 seats
- SUV — 6 seats
- **Result: 2 cars = 10 seats total**

2.4 Customer Panel — Feature List

- Sign up / login (phone number or email)
- Homepage with the 6 service options
- Automatic filtering: only cars registered near the selected pickup area are shown
- Car listing view: photos, model, seat capacity, rate, driver rating
- Booking confirmation with online advance payment
- "My Bookings" page showing status: Pending / Confirmed / Ongoing / Completed / Cancelled
- Option to rate and review the driver/car after a completed trip

⚠ Developer Notes — Booking Data

- Every service form submission creates a booking request in the backend containing: service type, all field data, customer info, and status (pending / confirmed / completed).
- Keep the pickup location field structured (dropdown or map-based area select), not free-text — this is required to filter and match the nearest registered car owner/driver.
- Intercity Travel is a standalone service, separate from Travel & Tour — Intercity is for straightforward point-to-point city-to-city trips, while Travel & Tour covers multi-destination/multi-day tour packages with an itinerary.

3. Owner / Driver Registration Panel

Step-by-step registration flow for car owners and rental businesses.

Step 1 — Basic Info

- Name (owner / business name)
- Phone number (OTP verification recommended)
- Email (optional)
- Business type: Individual owner / Rental agency (if multiple cars)
- Area / address (main operating zone — where pickup usually happens)

Step 2 — Owner/Business Verification Documents

- NID card (front-back photo upload)
- Trade license (if rental business / agency)
- Address proof (utility bill)

Step 3 — Car Information ("Add another car" option if multiple)

- Car name/model (e.g. Toyota Axio, Noah, Hiace)
- Manufacturing year (Optional)
- Seat count
- Car color
- Number plate
- Car photos (front, back, inside — a few)
- Registration paper (BRTA) upload
- Fitness certificate upload (Expire date entry must)
- Tax token upload
- Insurance paper (if available)

Step 4 — Driver Information (if owner keeps a driver instead of driving)

- Driver's name
- Phone number
- Driving license (photo upload, front-back)
- Driver's photo
- Experience (years, optional)

If the owner is the driver themself, this step is skipped.

Step 5 — Service Preference

- Which services they want to offer (checkbox: Airport Transfer / Corporate / Rental / Intercity / Tour / Event) (Choose any 3 options)
- Availability: Full-time / Part-time / Weekend only
- Operating area (select one or more areas)

Step 6 — Payment / Bank Info

- bKash/Nagad number (for receiving payouts)
- Bank account (optional, if bank transfer is kept)

Step 7 — Submit → Pending Approval

1. After submitting, status stays "Pending"
2. A notification goes to the admin panel
3. Admin verifies all documents and Approves/Rejects
4. On approval, the owner is notified via SMS/email — from then, their car becomes visible for customer booking

3.1 Owner's Own Dashboard (after approval)

- See booking requests for their car (notification on new booking)
- Option to accept/reject bookings
- See their booking history and earnings
- Update car info/photos
- If multiple cars, manage all cars (mark available/unavailable)
- Update profile/documents (e.g. upload a new fitness certificate)
- Earnings dashboard: total earned, commission deducted, net payable, payment history
- Availability calendar management

> ⚠ **Developer Notes — Registration & Verification**
>
> - For multi-car owners (rental agency), one owner account needs multiple car profiles under it (one-to-many relation).
> - Document upload should accept both image.
> - In MVP, driver info can be optional (can start with just owner+car verification) — driver management can be added later.
> - Must ensure the logic: without approval, a car should never show up in customer-facing search.

4. Admin Panel — Full Structure

Complete panel-by-panel design for the Rownaa admin — covering owner registration, bookings, drivers, payments, and platform control.

Panel 1 — Dashboard (Overview)

- Total bookings — today / this week / this month
- Total revenue and commission earned — today / this week / this month
- Pending bookings (not yet assigned to a driver)
- New owner/driver registration requests (pending approval)
- Active vs. completed vs. cancelled booking counts
- Quick alerts: expiring fitness certificates/tax tokens, expiring subscriptions

Panel 2 — Owner Registration Panel

Two-layer verification: the business/company is approved first, then each car under it is verified and approved separately.

2A. Registration Queue

- List of all registration submissions — filter by Pending / Approved / Rejected
- Filter by business type (Individual owner / Rental agency), area, and submission date
- Quick search by owner name or phone number

2B. Business/Company-Level Review

- Basic info: owner/business name, phone (OTP-verified), email, area/operating zone
- Verification documents: NID (front-back), trade license (if agency), address proof
- bKash/Nagad number and bank info (for payouts) shown for reference
- Approve / Reject button with a mandatory reason note on rejection
- On approval, the owner is notified via SMS/email; on rejection, the reason is included
- Option to suspend or block an already-approved business
- View total car count and status summary under this business

2C. Car-Level Review (per business)

- Every car added under an approved business appears here as its own entry ("Add Car")
- Car details: model, manufacturing year, seat count, color, number plate, photos (front/back/inside)
- Document verification per car: BRTA registration paper, fitness certificate, tax token, insurance (if available)
- Each car is individually Approved/Rejected — one car's rejected papers don't affect the business's other cars
- Car status: Active / Inactive / Suspended (auto notify us -suspend on fitness/tax token expiry, or manual)
- Driver assignment per car — pick a specific driver or from the business's driver pool
- Logic enforced platform-wide: an unapproved car never appears in customer-facing search

2D. Driver Review

- Driver list with NID, driving license (front-back), photo, years of experience
- Which car each driver is currently assigned to
- Individually approve, suspend, or block a driver
- In MVP, driver verification can be optional — owner-and-car verification is enough to launch

Panel 3 — Booking Panel

3A. Booking List & Detail

- Full booking list — filter by service type, status, date, area
- Booking detail view: customer info + all form data the customer submitted
- Manual driver/car assignment — system suggests matching owners based on the booking's pickup area

3B. Booking Status & Trip Control

- Status flow: Pending → Confirmed → Ongoing → Completed → Cancelled
- "Start Trip" — marks the booking Ongoing once the driver has picked up the customer
- "End Trip / Complete" — marks drop-off and completion; admin can view real-time trip status or override it if needed
- Completed trips automatically move into booking history and trigger the payment/commission step

3C. Payment on the Booking

- Advance payment (bKash/Bank or SSLCommerz) received status, shown directly on the booking
- "Bill Payment Receive" log — final amount entered by the owner/driver (base fare + extras such as waiting time or extra km)
- Payment method shown: Cash / bKash / Nagad / Bank Transfer
- Commission (10%) auto-calculated from the received amount and reflected in the admin's revenue report
- Cancel/refund handling directly from the booking detail view
- Booking history and rating viewable per car and per customer

Panel 4 — Payment & Commission

- Transaction log of every booking's advance payment (bKash/Bank Transfer or SSLCommerz)
- Commission calculation (10%) — how much commission each booking generated
- Owner payout tracking (for when Rownaa collects the full payment and pays owners out)
- Monthly/weekly revenue report

Panel 5 — Subscription Management

- Create/edit subscription plans (e.g. Free / Premium / Business) — car limits, visibility priority, commission rate per plan
- Manage subscription price, duration, benefits, and active/expired status
- View each owner's active/expired subscription status
- Renewal reminders sent via SMS/email before expiry
- That owner's vehicle(s) will automatically move to the regular category until the expired subscription is renewed.

Panel 6 — Service & Content Management

- Edit/add/remove the 6 service categories shown on the customer homepage
- Update homepage banner/promo content
- Edit pricing guideline or estimated fare, if used

Panel 7 — Customer Management

- Registered/guest customer list
- Booking history per customer
- Complaint/feedback management

Panel 8 — Notifications

- Admin notified (email/SMS) when a new booking comes in
- Notification when a new owner/driver approval request comes in
- Customer notified by SMS/email when a booking is confirmed and when the trip starts/completes

Panel 9 — Roles & Access (for later)

- Initially, a single admin account manages everything
- When staff are added later, role-based access applies (e.g. a booking manager sees only bookings, not payments)

5. Trip Lifecycle & Payment Workflow

5.1 Trip Start & Trip Complete

- “Trip Start” and “Trip Complete” buttons for each booking
- Automatically record date/time and update trip status on each action
- Trip Complete/End should only be available after Trip Start

5.2 Bill Payment Received

- "Payment Received" button becomes available after trip completion/end
- Show Total Bill, Received Amount, Due Amount, and Payment Status
- Payment status: Unpaid / Paid
- Save all payment records in the Admin Panel

5.3 Advance Payment & Commission

- Online advance payment (20% of fare) via bKash /Bank Transfer or SSLCommerz integration at checkout
- Remaining balance paid directly to the driver at the time of service
- Admin panel reconciles advance payments against commission owed per booking
- Commission (≈10%) auto-calculated from the received amount

6. Notes for the Developer — Summary

- Platform owner will not supply vehicles — all cars and drivers come from registered owners
- Owner/driver verification must be completed by admin before a listing goes live
- Location-based matching (pickup area → nearby registered cars) is a core requirement, not an optional filter
- Seat-matching logic should suggest car combinations automatically based on passenger count
- Mobile-responsive design is required since most customers are expected to book from phones
- Architecture should allow adding new service categories and coverage areas without a rebuild
- Keep the one business → many cars (one-to-many) relation in the database design from day one
- Document upload should accept both image and PDF formats
- In MVP, driver info/verification can be optional — owner + car verification is enough to launch
- Intercity Travel and Travel & Tour are separate services — do not merge their forms or logic

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ROWNAA — Car rental marketplace — book by location, drive with trust